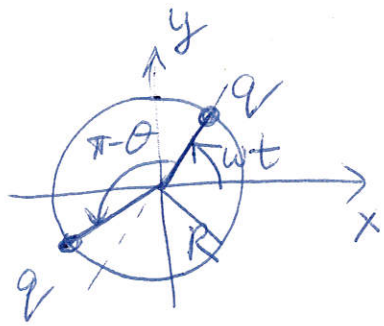
P1



$$\vec{d} = \sum_{j=1,2} q_j \vec{r}_j = qR\Big[\cos(\omega t)\hat{x} + \sin(\omega t)\hat{y} + \cos(\omega t + \pi - \theta)\hat{x} + \sin(\omega t + \pi - \theta)\hat{y}\Big]$$

(a)

$$\vec{d} = qR\Big\{\underbrace{\big[\cos(\omega t) - \cos(\omega t - \theta)\big]}_{\cos(\omega t)(1-\cos\theta) - \sin(\omega t)\sin\theta}\hat{x} + \underbrace{\big[\sin(\omega t) - \sin(\omega t - \theta)\big]}_{\sin(\omega t)(1-\cos\theta) + \cos(\omega t)\sin\theta}\hat{y}\Big\}$$

In order for electric dipole contribution to be as small as electric quadrupole, the angle $\theta \ll 1$
(for $\theta \sim 1$ dipole intensity is much stronger than quadrupole)
$\Rightarrow$ can approximate $\cos\theta \simeq 1$, $\sin\theta \simeq \theta$

$$\Rightarrow \vec{d} \cong qR\theta\Big\{-\sin(\omega t)\hat{x} + \cos(\omega t)\hat{y}\Big\}$$

(check: for $\theta = 0$ $\vec{d} = 0$)

To obtain $Q_{\alpha\beta}$ we can set $\theta = 0$ as $Q \neq 0$ when $\theta = 0$. This calculation was done in Problem 1 of Midterm 1 previously. The result

$$\hat{Q} = 2qR^2 \begin{pmatrix} 3\cos^2\omega t - 1 & 3\cos\omega t \sin\omega t & 0 \\ 3\cos\omega t \sin\omega t & 3\sin^2\omega t - 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\vec{Q}_\alpha = Q_{\alpha\beta} n_\beta \Rightarrow Q_x = 2qR^2\Big((3\cos^2\omega t - 1)n_x + 3\cos\omega t \sin\omega t\, n_y\Big)$$

$$Q_y = 2qR^2\Big((3\sin^2\omega t - 1)n_y + 3\cos\omega t \sin\omega t\, n_x\Big)$$

$$Q_z = -2qR^2 n_z$$

$$\dddot{\hat{Q}} = \frac{d^3}{dt^3}\left\{ 3qR^2 \begin{pmatrix} \cos(2\omega t) & \sin(2\omega t) & 0 \\ \sin(2\omega t) & -\cos(2\omega t) & 0 \\ 0 & 0 & 0 \end{pmatrix} + \text{const} \right\} =$$

$$= 3qR^2 (2\omega)^3 \begin{pmatrix} \sin(2\omega t) & -\cos(2\omega t) & 0 \\ -\cos(2\omega t) & -\sin(2\omega t) & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\dddot{Q}_x = (24 q R^2 \omega^3)\left(\sin(2\omega t)\, n_x - \cos(2\omega t)\, n_y\right)$$

$$\dddot{Q}_y = (24 q R^2 \omega^3)\left(-\cos(2\omega t)\, n_x - \sin(2\omega t)\, n_y\right)$$

From lectures (and Brau's book, Ch. 10)

$$p^{(EQ)} = \frac{\mu_0}{720\pi c^3} \sum_{\alpha\beta} \dddot{Q}_{\alpha\beta}^2 = \frac{\mu_0}{720\pi c^3}\, 2(24 q R^2 \omega^3)^2 = \frac{8\mu_0 q^2 R^4 \omega^6}{5\pi c^3}$$

$$\hookrightarrow \dddot{Q}_{xx}^2 + \dddot{Q}_{xy}^2 + \dddot{Q}_{yx}^2 + \dddot{Q}_{yy}^2 =$$

$$= (24 q R^2 \omega^3)^2 (2)\left(\sin^2(2\omega t) + \cos^2(2\omega t)\right)$$

Now electric dipole $\ddot{\vec{d}} = qR\theta\omega^2\left(\sin(\omega t)\hat{x} - \cos(\omega t)\hat{y}\right)$

$$p^{(ED)} = \frac{\mu_0}{6\pi c} |\ddot{\vec{d}}|^2 = \frac{\mu_0}{6\pi c}(qR\theta\omega^2)^2 = \frac{\mu_0 q^2 R^2 \omega^4 \theta^2}{6\pi c}$$

$$p^{(ED)} = p^{(EQ)} \Rightarrow \frac{\mu_0 q^2 R^2 \omega^4 \theta^2}{6\pi c} = \frac{8\mu_0 q^2 R^4 \omega^6}{5\pi c^3}$$

$$\theta^2 = \frac{48}{5}\frac{R^2\omega^2}{c^2}$$

$$\theta = 2\sqrt{\frac{12}{5}}\frac{R\omega}{c} \ll 1 \text{ for } R\omega \ll c.$$

Problem 2

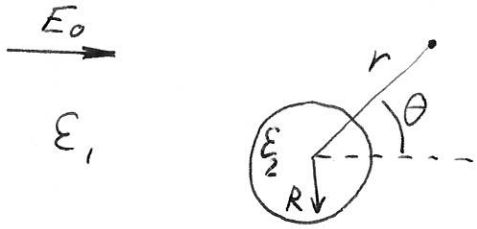
a) Displacement $\vec{D}$ obeys the equation

$$\nabla \cdot \vec{D} = \rho_{free}$$

which does not depend on ε_1 at all:

$$D = \sigma_0 \quad \Rightarrow \quad \varepsilon_1 E_0 = \sigma_0 \quad \Rightarrow \quad \boxed{E_0 = \frac{\sigma_0}{\varepsilon_1}} \qquad (1)$$

b) Since the radius of the cylinder is small ($R \ll d$) far away from the cavity the field is given by the answer to part a). That will serve as one of the boundary conditions:



$$\varphi = -E_0 r \cos\theta \quad , \quad r \to \infty$$

Laplace equation in cylindrical coordinates:

$$\nabla^2 \varphi = 0 \quad \Rightarrow \quad \frac{1}{r}\frac{\partial}{\partial r}\left(r \frac{\partial \varphi}{\partial r}\right) + \frac{\partial^2 \varphi}{r^2 \partial^2 \theta} = 0 \qquad (2)$$

We look for a solution, $\sim \cos\theta$, suggested by the boundary condition

$$\varphi = f(r)\cos\theta$$

where $f(r)$ obeys

$$f'' + \frac{1}{r} f' - \frac{f}{r} = 0 \qquad (3)$$

General solution has the form

$$f(r) = Ar + \frac{B}{r} \qquad (4)$$

Inside $B = 0$:

$$f(r<R) = Ar \tag{5}$$

Outside $A = -E_0$

$$f(r) = -E_0 r + \frac{B}{r} \tag{6}$$

The boundary conditions at $r=R$ require the continuity of φ:

$$AR = -E_0 R + \frac{B}{R} \tag{7}$$

(this is the same condition as required by the continuity of $\vec{E}_{\parallel}$), and the continuity of D_r, where $D_r = -\varepsilon\, \partial\varphi/\partial r$

$$\varepsilon_2 A = \varepsilon_1 (-E_0) - \varepsilon_1 \frac{B}{R^2} \tag{8}$$

Multiplying Eq. (7) by $\frac{\varepsilon_1}{R}$ and adding the result to Eq. (8) we get

$$(\varepsilon_1 + \varepsilon_2) A = -2 E_0 \varepsilon_1 \quad \Rightarrow \quad A = -\frac{2\varepsilon_1}{\varepsilon_1 + \varepsilon_2} E_0$$

From Eq. (7) now:

$$B = -\frac{\varepsilon_1 - \varepsilon_2}{\varepsilon_1 + \varepsilon_2} E_0 R^2$$

This gives potential inside

$$\boxed{\varphi_{in} = -\frac{2\varepsilon_1}{\varepsilon_1 + \varepsilon_2} \vec{E}_0 \cdot \vec{r}} \tag{9}$$

Again, E_0 is given by Eq. (1).

which corresponds to the uniform field $2\varepsilon\vec{E}_0/(\varepsilon_1+\varepsilon_2)$.

The potential outside:

$$\boxed{\varphi_{out} = -\vec{E}_0\cdot\vec{r} - \frac{\varepsilon_1-\varepsilon_2}{\varepsilon_1+\varepsilon_2}\,\frac{\vec{E}_0\cdot\vec{r}}{r^2}} \qquad (10)$$

Electric field in components can be written from Eq. (10), $\vec{E} = -\nabla\varphi$:

$$E_r = -\frac{\partial\varphi}{\partial r} = E_0\cos\theta - \frac{\varepsilon_1-\varepsilon_2}{\varepsilon_1+\varepsilon_2}\,\frac{E_0\cos\theta}{r^2}$$

$$E_\theta = -\frac{1}{r}\frac{\partial\varphi}{\partial\theta} = -E_0\sin\theta - \frac{\varepsilon_1-\varepsilon_2}{\varepsilon_1+\varepsilon_2}\,\frac{E_0\sin\theta}{r^2}$$

c) The electrostatics approach is valid as long as the second term in the equation for scalar potential

$$\nabla^2\varphi + \frac{\omega^2}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0}\varphi = 0$$

is small compared with the first one.

$$\nabla^2\varphi \sim \frac{\varphi}{d^2}, \text{ while } \frac{\omega^2}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0}\varphi = \frac{\varphi(\omega_0^2-\omega^2)}{c^2} \sim \frac{\varphi\omega_0^2}{c^2}$$

so that we need

$$\frac{\omega_0}{c} \ll d^{-1} \quad \text{or} \quad d/c \ll \omega_0^{-1}$$

d) Dissipation occurs only inside the metal where under the condition of part c)

the field is given by straightforward generalization of Eqs. (1) and (10):

$$E(t) = E_\omega e^{-i\omega t} + E_\omega^* e^{i\omega t} \qquad (11)$$

where $E_\omega = \dfrac{z_0}{\varepsilon_1 + \varepsilon_2(\omega)}$ and $\varepsilon_2^*(\omega) = \varepsilon_2(-\omega)$

Drude conductivity is $z(\omega) = \dfrac{ie^2 n}{m(\omega + i/\tau)} = i\varepsilon_0 \dfrac{\omega_p^2}{\omega + i/\tau}$

Electric current from (11) is

$$j(t) = E_\omega z(\omega) e^{-i\omega t} + E_\omega^* e^{i\omega t} z^*(\omega) \qquad (12)$$

(Do not confuse $z(\omega)$ with z_0!)

Total dissipated power is

$$P/V = \overline{j(t) \cdot E(t)} = |E_\omega|^2 \left(z(\omega) + z^*(\omega) \right)$$

$$z(\omega) + z(-\omega) = \varepsilon_0 \omega_p^2 \left(\frac{i}{\omega + i/\tau} - \frac{i}{\omega - i/\tau} \right) = \frac{2\varepsilon_0 \omega_p^2 \tau^{-1}}{\omega^2 + \tau^{-2}}$$

$$|E_\omega|^2 = \frac{z_0^2}{|\varepsilon_1 + \varepsilon_2(\omega)|^2} = \frac{z_0^2}{\left| \varepsilon_1 + \varepsilon_0 - \varepsilon_0 \dfrac{\omega_p^2}{\omega(\omega + i\tau^{-1})} \right|^2} =$$

$$= \frac{z_0^2 \omega^2 (\omega^2 + \tau^{-2})}{(\varepsilon_1 + \varepsilon_0)^2 \left[(\omega^2 - \omega_0^2)^2 + \omega^2 \tau^{-2} \right]}$$

where $\omega_0^2 = \dfrac{\varepsilon_0 \omega_p^2}{\varepsilon_1 + \varepsilon_0}$ is the resonance frequency

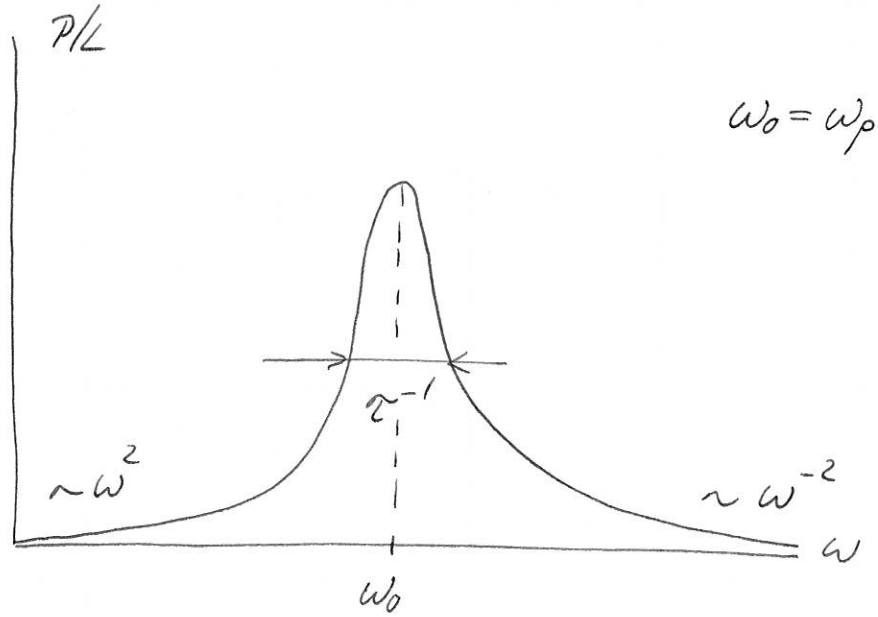
The volume $V = \pi R^2 L$.

This gives dissipated power per unit length:

$$P/L = \pi R^2 \times \frac{z_0^2 \omega^2 (\omega^2 + \tau^{-2})}{(\varepsilon_1 + \varepsilon_2)^2 \left[(\omega^2 - \omega_0^2)^2 + \omega^2 \tau^{-2} \right]} \times \frac{2 \varepsilon_0 \omega_p^2 \tau^{-1}}{\omega^2 + \tau^{-2}} =$$

$$= \frac{2 \pi R^2 z_0^2 \omega^2 \omega_p^2 \varepsilon_0 \tau^{-1}}{(\varepsilon_1 + \varepsilon_2)^2 \left[(\omega^2 - \omega_0^2)^2 + \omega^2 \tau^{-2} \right]} \qquad (13)$$

e)



$$\omega_0 = \omega_p \sqrt{\frac{\varepsilon_0}{\varepsilon_1 + \varepsilon_0}}$$

The resonance is sharp as long as

$$\tau^{-1} \ll \omega_0 .$$